

ANIMAL GROUPS AND NAMES

The following 5 pages summarize the classification system used in this book. Like all classification systems, it represents current thinking, which is always liable to change. Groups are “nested” to show how they are related; informal groups with no distinct biological identity are bounded by dotted lines. Some of the species totals, particularly for the invertebrates, are based on estimates; as with new discoveries, these can be subject to rapid change.

Animal groups

The animal kingdom is divided into more than 30 phyla. Some of the smallest phyla contain less than 100 species, while at the other extreme, the arthropod phylum contains as least 1.2 million, making these arguably the most successful animals of all time. Between these 2 extremes are the chordates—animals that usually have a backbone, and that make up a large part of this book.

Although not front runners in terms of species, chordates are exceptional in many ways. They show an extraordinary range in dimensions, from fish the size of a fingernail to majestic 100-ton whales. They can be found on land, at sea, and in the air, where they outrun, outswim, or outfly all other forms of life. Since antiquity, they have been much observed, which is why many names for animal groups have their origin in those times.

Group names

Like species names, group names often look obscure but they have a language and logic of their own. Many of them are named after particular body parts that make a group distinct. Chordates, for example, get their name from the notochord—a strengthening rod that runs down their backs for part or all of their lives. Arthropods get their name from their joint-bearing legs, one of the features that accounts for their success. Among chordates, frogs and toads make up the order Anura, which simply means “without tails,” while elephants make up the order Proboscidea, after their long proboscis, or nose.

Sometimes, ancient folklore also plays a part in names. For example, nightjars and frogmouths, in the avian order Caprimulgiformes, got their name from the belief that nightjars sucked milk from goats. Dugongs and manatees are more fanciful still. They belong to the order Sirenia, from their supposed resemblance to the three sirens of Greek mythology—mythical figures who seduced sailors into shipwreck on the rocks.

An ongoing process

Classification changes all the time as more is discovered about animals and their evolutionary past. Some changes affect life at its highest levels: today, for example, it is generally accepted that there are 5 kingdoms of life (see p.14), but in the recent past there were 6. Changes to lower-level classification happen every day. Through cladistics and DNA analysis, much more information about the relationships between species is becoming available, and as a result, groups at different taxonomic levels are often moved, split, or joined together.

Seals, for example, were formerly considered to be a separate group of mammals, the order Pinnipedia. Today, most zoologists classify them as a family within the order Carnivora, together with terrestrial carnivores such as cats, dogs, and bears. One widely accepted change not reflected here is the creation of the order Cetartiodactyla, which shows the close relationships between even-toed hoofed animals and the cetaceans. With moves like these, the animals themselves do not change. What does change is our understanding of them: how they have evolved, and which animals are their closest relatives.

VERTEBRATES



MAMMALS

CLASS Mammalia

GREVY'S ZEBRA

Mammals are chordates characterized by having fur or hair, and raising their young on milk. There are over 5,000 living species, in 29 orders, ranging from a single species to over 2,000. Mammals are divided into 3 groups, depending on their reproductive strategy: monotremes, marsupials (Metatheria), and placental mammals (Eutheria). Monotremes lay eggs, while marsupials and placentals give birth to live young. Marsupial young are born at an early stage, and complete development in a pouch. In placentals, young are nourished inside their mother's body, so they are at a more advanced stage at birth. Even so, parental care may last at least a year.

EGG-LAYING MAMMALS

ORDER Monotremata	FAMILIES 2	SPECIES 5
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MARSUPIALS

INFRAClass Marsupialia	FAMILIES 19	SPECIES 363
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AMERICAN OPOSSUMS

ORDER Didelphimorphia	FAMILIES 1	SPECIES 103
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AUSTRALASIAN CARNIVOROUS MARSUPIALS

ORDER Dasyuromorphia	FAMILIES 2	SPECIES 75
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BANDICOOTS

ORDER Peramelemorphia	FAMILIES 3	SPECIES 19
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MARSUPIAL MOLES

ORDER Notoryctemorphia	FAMILIES 1	SPECIES 2
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KANGAROOS AND RELATIVES

ORDER Diprotodontia	FAMILIES 11	SPECIES 156
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SHREW OPOSSUMS

ORDER Paucituberculata	FAMILIES 1	SPECIES 6
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MONITO DEL MONTE

ORDER Microbiotheria	FAMILIES 1	SPECIES 1
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SENGIS

ORDER Macroscelidea	FAMILIES 1	SPECIES 15
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TENRECS AND GOLDEN MOLES

ORDER Afrosoricida	FAMILIES 2	SPECIES 51
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AARDVARK

ORDER Tubulidentata	FAMILIES 1	SPECIES 1
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DUGONG AND MANATEES

ORDER Sirenia	FAMILIES 2	SPECIES 4
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ELEPHANTS

ORDER Proboscidea	FAMILIES 1	SPECIES 3
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HYRAXES

ORDER Hyracoidea	FAMILIES 1	SPECIES 5
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ARMADILLOS

ORDER Cingulata	FAMILIES 1	SPECIES 21
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SLOTHS AND ANTEATERS

ORDER Pilosa	FAMILIES 4	SPECIES 10
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RABBITS, HARES, AND PIKAS

ORDER Lagomorpha	FAMILIES 2	SPECIES 92
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